

Positioning for Risk-Off: A Methodology for Sovereign Credit Ratings

Methodology document, v0.5 — for discussion

Alex Stomper, HU Berlin

August 2026

Abstract

This document specifies a sovereign credit rating methodology organized around a single idea: *countries differ in how they are positioned for risk-off — episodes in which the market’s willingness to bear risk falls — and this positioning determines sovereign credit risk*. The underlying theory (Stomper 2026, “the Note”) delivers three regimes — a good equilibrium, a bad equilibrium, and a multiple-equilibrium (fragile) region — and measures a country’s position by its distances to the “exits” from its current regime, denominated in units of an estimable global risk-appetite factor. The methodology translates this into a transparent sequence of rating tasks: a *fundamentals block* that estimates a country’s growth distribution and fiscal capacity; a *state-of-affairs (SOA) block* that measures borrowing needs, borrowing costs, and the market’s risk appetite at monthly frequency; and a *scoring and rating block* that maps regime membership and exit distances into rating classes. Judgment enters the process only by calibrating documented inputs or through the published outlook — never as a direct adjustment of a score. All metrics rely on IMF data, freely available public sources, and (at most monthly) sovereign bond yields; every score is a deterministic, replicable function of a published parameter vector. Each block distinguishes *basic tasks* — which together constitute a minimal rating process (MRP), suitable for teaching and for a defensible first pass at any country — from *further tasks* that refine them. A proof-of-concept section applies the MRP, end to end, to Austria as of August 2026. Appendix A relates the approach, standard by standard, to benchmarks set by international organizations and academic publications; Appendix B summarizes the MRP as a two-page recipe.

1 Philosophy and the claim of the methodology

A sovereign rating is an ordinal opinion about the probability of default over a horizon. Existing methodologies form that opinion by scoring fundamentals — institutional, economic, fiscal, external, monetary — and mapping the scores to a letter grade. The premise of this methodology is different: for market-access sovereigns, near- and medium-term default risk is dominated by *where the country sits relative to a fold in its market access*, and by *how far global risk appetite would have to move* to push it over that fold. Two countries with identical debt ratios can carry very different risk because their funding curves respond differently to the same global repricing; conversely, a country can be violently repriced with no fiscal news at all. A rating process that scores fundamentals but not positioning will systematically miss the difference.

Throughout the document, “risk-off” has a precise meaning: a fall in the market’s willingness to bear risk, formalized as an *increase* in the market price of distress risk θ introduced in Section 3; “risk-on” is the opposite move. The theory behind the methodology’s claim is stated in the Note and summarized in Section 3. Its objects are deliberately spartan: a country is characterized by a small parameter vector — maximum primary surplus $\bar{s}$, growth distribution (μ, σ) , safe rate R^f , haircut h — and a short-run state consisting of the refinancing need n_t (in practice: the published gross financing need) and the market price of distress risk θ_t (in practice:

a monthly estimate from global market data). From these the model computes the maximum sustainable borrowing b_M , the band edges $n_L(\theta_t)$, $n_U(\theta_t)$ of the multiple-equilibrium region, and, for each country, two *distances to the exits*:

$$\Delta_G = \theta_G(n_t) - \theta_t \quad (\text{the risk-off move that would destroy the good equilibrium}),$$

$$\Delta_B = \theta_t - \theta_B(n_t) \quad (\text{the risk-on move that would destroy the bad one}),$$

together with a well-defined tail probability, $\text{Exposure}_{it} = \mathbb{P}[\theta > \theta_G(n_{it})]$: the probability of a risk-appetite realization that leaves the country with only the bad equilibrium. The rating, in this methodology, is a coarse public signal about these coordinates.

Three principles follow and are maintained throughout.

1. Rate the coordinates, not the price. Over the fragile region two market spreads are consistent with the same fundamentals; the spread cannot be inverted into fundamentals there. A country repriced from the good to the bad equilibrium on a pure risk-appetite move has not changed coordinates and is not mechanically downgraded; a country enjoying good-equilibrium pricing while drifting toward the upper band edge *is* downgraded before the spread moves. Ratings and market prices measure different objects, and the methodology says exactly which.

2. Positioning is measured in units of a global, estimable factor. Distances Δ_G , Δ_B are denominated in units of the market price of distress risk θ — the same units for every country — and are convertible into probabilities given the estimated distribution of the factor. This is what makes ratings comparable across countries: not a common checklist, but a common state variable.

3. Transparency means replicability. Every rating is a deterministic function of a published parameter vector $(\bar{s}, \mu, \sigma, R^f, h; n_t, \theta_t)$. Given the published inputs and the public code, any user reproduces the rating exactly. This is the standard set by academic replication practice and by open-data initiatives such as econdataverse, and it is deliberately stronger than the transparency practice of the existing standards reviewed in Appendix A.

4. Discretion calibrates inputs or informs the outlook; it never moves a score. The process contains judgment, but confines it to two channels. *Soft information* — the political economy of the maximum surplus, who holds the debt, backstop politics — may inform the *inputs* of the otherwise non-discretionary process: a parameter, a function, a distribution, always within documented ranges and with published justification, so that its effect on the rating flows through the same equations as every other input. And *information beyond the model's structure* may inform the *outlook* published alongside the rating — discretionary in the precise sense that it draws on facts the model does not represent, while respecting the model's purpose as a framework for thinking about a well-defined form of fragility. What judgment may never do is move a score or a rating class directly: the methodology has no notches. A notch is an untraceable displacement of the output — it severs the link between published inputs and published rating that Principle 3 establishes, and it reintroduces exactly the opacity the methodology exists to remove. Confining discretion to inputs also disciplines it: an analyst who disagrees with a rating must identify the input they disbelieve and defend an alternative value, rather than adjust the conclusion.

2 Scope

The methodology targets **advanced market-access sovereigns**. Some of these sovereigns benefit from a *backstop*: a commitment by an institution whose resources the market cannot

exhaust — in practice a central bank, or an official lender such as the ESM — to purchase or guarantee the sovereign’s debt under stated conditions. A credible backstop matters in this methodology because it removes the self-fulfilling refusal to lend: if lenders know a deep-pocketed buyer stands behind the debt, the belief “others will not lend” can no longer justify not lending, and the bad equilibrium disappears from the equilibrium set (Section 3). Within the scope, two tiers receive different treatment of the monetary pillar (Section 5):

- **Monetary-union members (euro area): full application.** These sovereigns borrow in a unit of account they cannot print (the Bolton–Huang premise), so the fragile-region apparatus applies with full force. The safe rate R^f is the union-wide curve; the backstop (ECB/ESM) is institutionally separate from the sovereign, is conditional, and its *eligibility criteria* become a rating input. The euro area receives special treatment as the home terrain of the theory: the SOA block runs on union-wide factor estimates, and cross-country comparability of exit distances is exact because all members face the same θ_t .
- **Own-currency sovereigns (e.g., US, UK, Japan): modified application.** A monetarily sovereign issuer cannot be forced into *nominal* default by a run; the backstop is constitutionally built in, which deletes the bad equilibrium for nominal claims. The fundamentals block still binds — d_M is finite — but the fragile-region metrics are reported as *shadow* quantities, and the residual risk channel (inflation as the alternative to an explicit haircut) is flagged separately rather than folded into the default rating. Section 5 specifies the modification.

Emerging market-access economies can be rated with the same pipeline provided their debt is predominantly in domestic currency or the FX-denominated share is modeled as a separate seniority class; this extension is noted where relevant but not developed here. Low-income countries under the IMF–World Bank LIC-DSF are out of scope.

3 The theoretical core, restated compactly

This section restates the Note so that the document is self-contained: every variable and function used later is defined here or at its point of first use. Two notational conventions. First, all rates and growth factors are *gross*: $R^f = 1.02$ means a safe rate of 2 percent, $G_{t+1} = 1.03$ means nominal growth of 3 percent. Second, $\Phi(\cdot)$ denotes the cumulative distribution function (cdf) of the standard normal distribution and $\phi(\cdot)$ its density; these are the only distribution functions used, apart from the growth cdf F defined below, which is itself expressed through Φ .

Primitives. Time is annual. Nominal GDP y_t grows at the gross rate $G_{t+1} = y_{t+1}/y_t$, independently and identically distributed (i.i.d.) lognormally:

$$\ln G_{t+1} \sim N(\mu, \sigma^2), \quad F(x) = \mathbb{P}[G \leq x] = \Phi\left(\frac{\ln x - \mu}{\sigma}\right), \quad \bar{g} = \mathbb{E}[G] = e^{\mu + \sigma^2/2}. \quad (1)$$

The gross safe return R^f is set outside the country (for a monetary-union member, by the union-wide safe asset). The government can sustain a primary surplus of at most $\bar{s}$ per year (as a share of GDP) and follows a fiscal plan $\{s_t\}$ with $s_t \leq \bar{s}$. Sovereign debt is a one-period zero-coupon bond: d_t denotes the face value due at $t+1$ per unit of date- t GDP. In default, face values are written down by the haircut $h \in (0, 1)$; the recovery per unit of face value is $1 - h$. Lenders are competitive.

The refinancing need. At date t , growth G_t realizes, the inherited obligation per unit of current GDP is d_{t-1}/G_t , the government runs the surplus s_t , and must raise at auction the **refinancing need**

$$n_t = \frac{d_{t-1}}{G_t} - s_t, \quad (2)$$

an accounting identity. With a maturity structure, n_t is maturing debt plus interest minus the primary surplus — the *gross financing need* (GFN), a published statistic. This observability matters for everything downstream: the model’s short-run state variable is not a latent construct but a number national treasuries and official DSAs publish.

Pricing a defaultable promise. Let p denote the probability that the bond issued at t defaults at $t+1$. A lender indifferent between the safe asset and a bond that pays 1 with probability $1 - p$ and the recovery $1 - h$ with probability p prices a unit of face value at $q(p) = (1 - hp)/R^f$; the gross promised return $R = 1/q$ and the **spread** ψ follow:

$$R = \frac{R^f}{1 - hp}, \quad \psi = R - R^f = \frac{R^f h p}{1 - hp} \approx h p \quad \text{for small } hp. \quad (3)$$

The funding curve and the three regimes. Suppose the default probability at $t+1$ depends on the face value issued at t through a nondecreasing function $P : [0, \infty) \rightarrow [0, 1]$ with $P(0) = 0$ (its construction from fundamentals follows below). The auction raises, per unit of face value offered d , the price $q(P(d))$; total proceeds define the **funding curve**

$$\xi(d) \equiv \frac{d(1 - hP(d))}{R^f}, \quad (4)$$

and an equilibrium of the date- t auction is a solution d of $\xi(d) = n_t$. Because lenders demand compensation for expected write-downs, proceeds are not monotone in face value offered wherever P rises steeply: ξ can *fall* as d rises. Formally, ξ has a local maximum followed by a local minimum — an interior “dip” — if and only if

$$h[P(d) + dP'(d)] > 1 \quad \text{for some } d. \quad (5)$$

We refer to (5) as the *dip condition*; it holds whenever the haircut is material and P has a sufficiently steep region. Under it, denote by $d_U < d_L$ the two interior stationary points of ξ (the solutions of $\xi'(d) = 0$) and by

$$n_U = \xi(d_U), \quad n_L = \xi(d_L) \quad (6)$$

the proceeds at the local maximum and the local minimum. Three regimes result. For $n_t < n_L$ the equilibrium is unique with low p : the country is **safe**. For $n_L < n_t < n_U$ three equilibria coexist — a good one G (low spread), an unstable one, and a bad one B (spread up to the recovery ceiling $R^f h/(1 - h)$, by (3) at $p \rightarrow 1$) — and which prevails is selected by beliefs, not fundamentals: the country is **fragile**. For $n_t > n_U$ only the bad equilibrium remains: the country is **distressed**.

The default-probability function from fundamentals. Default at $t+1$ occurs if and only if growth comes out too low for the refinancing need to be coverable given **maximum sustainable borrowing** b_M — the largest amount the market will lend against future surpluses and rollovers (Collard–Habib–Rochet). Under (1),

$$P(d) = F\left(\frac{d}{\bar{s} + b_M}\right), \quad b_M = \frac{\bar{s}\gamma}{R^f - \gamma}, \quad \gamma = \max_{\chi \geq 0} (1 - F(\chi))\chi \leq \bar{g}, \quad (7)$$

where χ is a growth threshold (default occurs when G falls below the threshold implied by the promised face value) and γ — the **borrowing factor** — is a default-risk-discounted certainty-equivalent growth factor. The maximum face value consistent with market access is $d_M = (\bar{s} + b_M)\chi_M$, with χ_M the maximizer in (7); d_M is “the cliff.” Because $\gamma < R^f$ at every relevant calibration — *including* when $R^f < \bar{g}$ — the capacity b_M is finite: the default discount restores contraction exactly where the naive Gordon-growth logic fails.

Systematic risk and the market price of distress risk. Sovereign default happens disproportionately in global bad states, so default risk is priced with a premium. This is modeled by a one-factor pricing kernel loading on standardized log growth $z = (\ln G - \mu)/\sigma$ with **market price of distress risk** $\theta \geq 0$: under the pricing (risk-neutral) measure, log growth has drift $\mu - \theta\sigma$ instead of μ , and *every formula above goes through with μ replaced by $\mu - \theta\sigma$* . The parameter θ_t is the theory’s representation of global risk appetite: θ_t high means risk-off, θ_t low means risk-on. Two facts about θ_t organize the operational process:

- **θ_t is time-varying and must be estimated.** It is not directly observable; it is a price, revealed by how risky assets are priced globally. Task B3 of the SOA block specifies its estimator $\hat{\theta}_t$ and the mapping from observable market data into the kernel units used here. The estimate is refreshed *monthly*: each month delivers a new $\hat{\theta}_t$, hence a new country coordinate $(n_t, \hat{\theta}_t)$ and, through the equations below, a new set of band edges and distances. The fundamentals block, by contrast, is re-estimated annually.
- **Persistence matters more than level.** The current θ_t reverts to a long-run anchor θ_∞ ; investors’ conjecture about the future price relevant for continuation borrowing capacity is $\theta_t^c = \theta_\infty + \rho_\theta(\theta_t - \theta_\infty)$, where $\rho_\theta \in [0, 1]$ measures how permanent the market believes the current level to be. θ enters the model twice — in the one-period drift (small effect) and in the continuation capacity $b_M(\theta_t^c)$ (large effect) — so *a risk-off episode is dangerous in proportion to how permanent the market believes it to be*. The MRP sets $\rho_\theta = 1$ (risk-off treated as permanent), the conservative choice; further task B3⁺ estimates ρ_θ .

Band edges, critical prices, distances. Write ξ_θ for the funding curve (4) evaluated at the risk-adjusted objects (drift $\mu - \theta\sigma$, capacity $b_M(\theta^c)$), and $n_L(\theta)$, $n_U(\theta)$ for the corresponding proceeds levels (6). Both edges fall as θ rises: risk-off shifts the whole funding curve down. For a country with need n , the **critical risk prices** are defined by inverting the edges,

$$\theta_G(n) : n_U(\theta_G) = n, \quad \theta_B(n) : n_L(\theta_B) = n; \quad (8)$$

the good equilibrium exists iff $\theta_t \leq \theta_G(n)$, the bad iff $\theta_t \geq \theta_B(n)$, and the country is fragile iff $\theta_t \in [\theta_B(n), \theta_G(n)]$. The **exit distances** are

$$\Delta_G = \theta_G(n_t) - \theta_t, \quad \Delta_B = \theta_t - \theta_B(n_t), \quad (9)$$

and the **exposure probability** is the chance of a risk-appetite realization that leaves the country with only the bad equilibrium,

$$\text{Exposure}_{it} = \mathbb{P}[\theta > \theta_G(n_{it})], \quad (10)$$

computed under the distribution of θ specified in Task B3 (in the MRP, a normal distribution, so that (10) has a closed form in Φ).

Hysteresis and backstops. Once in B , returning to G requires not $\theta_t < \theta_G$ but the much stronger $\theta_t < \theta_B(n_t)$ — or a credible backstop announcement (Section 2), which deletes B from the equilibrium set without any purchases. Rating upgrades out of distress are therefore governed by a different, tighter condition than the downgrades into it, and backstop credibility is a first-class rating input.

4 The rating process

The process runs in three blocks. **Block A, the fundamentals block** (annual, with event-triggered re-runs), estimates the country’s parameter vector and computes its borrowing capacity

Discretionary input	Task (location)	Category	What it informs
placement of $\bar{s}$	A2 ⁺ (Block A)	calibration	fiscal capacity entering (13)
placement of h	A4 ⁺ (Block A)	calibration	haircut band position; width of $[n_L, n_U]$
factor loading β_i	B4 ⁺ (Block B)	calibration	country’s sensitivity to the global factor
eligibility state	DA2 (in C2)	calibration	factor distribution in (10); C3 upgrade gate
outlook	DA1 (in C4)	discretionary outlook	trajectory statement published with the rating

Table 1: Where discretion lives: five inputs, two categories, no notches.

— the country’s cliff. **Block B, the state-of-affairs (SOA) block** (monthly), measures the country’s coordinate: the refinancing need, borrowing costs, and the market’s risk appetite. **Block C, the scoring and rating block** (monthly, whenever Block B updates), classifies the regime, converts distances into rating classes, applies the two discretionary-adjustment (DA) tasks that live there — DA2, the backstop-eligibility component of the classification input, and DA1, the discretionary outlook — and publishes the rating. The design maps the course’s pillar table into an ordered workflow: growth and fiscal capacity first, borrowing needs and costs second, scoring last.

The discretionary layer. Principle 4 (Section 1) confines judgment to two categories, and the process locates each piece of discretion where the object it concerns lives (Table 1). *Calibration tasks* provide soft information for setting inputs of the otherwise non-discretionary process — a parameter ($\bar{s}$ in A2⁺, h in A4⁺), a function (the country’s loading on the global factor in B4⁺), a distribution (the eligibility-truncated factor distribution in DA2). Their effect on the rating flows through the published equations, so the audit trail shows precisely what the judgment did. *Discretionary outlook tasks* (DA1) take into account information beyond the model’s structure and inform only the outlook, never the class. Nothing in either category touches a score directly.

Three design rules govern all data choices. First, *IMF data by default*: the WEO, Fiscal Monitor, and the Arslanalp–Tsuda investor-base dataset cover most inputs; the only non-IMF essentials are bond yields (ECB SDW, FRED, national debt-management offices — all free, at most monthly) and, for EU members, the Darvas–Welslau–Zettelmeyer (DWZ) replication repository of the Commission DSA. Second, *easy econometrics only*: sample moments, ordinary least squares, an AR(1) fit, one principal component, and one-dimensional numerical solves; the entire pipeline runs in R or Python on free data. Third, *econdataverse alignment where it exists*: the econdataverse tool suite provides open R/Python packages for sovereign-debt data; the reference implementation uses them where coverage exists and falls back to direct API pulls where it does not — the initiative is the delivery mechanism, not the definition of any metric.

Basic and further tasks: the minimal rating process. Within each block, tasks are split into **basic** and **further** versions. The basic tasks of Blocks A, B, and C together constitute a complete, self-contained rating process — the *minimal rating process* (MRP) — with three properties: every input is a single published number or a sample moment; every computation is a closed form or a one-dimensional solve; and the whole process fits on one page (Table 2) and in one classroom session. A further task never introduces a new rating concept; it modifies a named output of a named basic task — a better estimator, a finer frequency, an uncertainty quantification — and each further task below states explicitly which output it modifies. Every basic task that computes something is stated as a numbered equation, so that any application of the MRP can cite, step by step, the exact expression it evaluates. The proof of concept in

Task	Output	Basic version, in one line
A1	$\hat{\mu}, \hat{\sigma}$	sample moments of log nominal GDP growth, last 25 years (IMF WEO)
A2	$\bar{s}$	best sustained 5-year average primary balance in the country's history
A3	R^f	the 1-year point of the euro-area AAA curve (own curve for own-currency tier)
A4	h	fixed at 0.30 (center of the historical haircut distribution)
A5	γ, b_M, d_M	closed forms (13), at $\theta^c = \hat{\theta}_t$ with $\rho_\theta = 1$
B1	n_t	GFN $\approx$ MLT amortizations + short-term stock + deficit, eq. (15)
B2	ψ_t	monthly 10-year spread over the safe curve; default component by (3)
B3	$\hat{\theta}_t$; θ -dist.	VIX-based plug-in estimator (16)–(17); $\theta \sim N(\theta_\infty, \sigma_\theta^2)$
B4	regime	solve (6) for $n_L(\hat{\theta}_t), n_U(\hat{\theta}_t)$; classify by n_t ; branch from spread
B5	distances	θ_G, θ_B by (8); Δ_G, Δ_B by (9); Exposure by (10); X^{fisc} by (18)
C1	regime class	confirm S/F/D assignment and prevailing branch
C2	class; DA2 state	threshold tables on Exposure, X^{fisc} , $\mathbb{P}[\theta < \theta_B]$; eligibility bright lines
C3	upgrade gate	hysteresis rule for exits from D (conditions on the DA2 state)
C4	outlook (DA1)	projected Δ_G direction along the n -path + beyond-model information
C5	publication	one-page parameter sheet + sensitivity grid

Table 2: The minimal rating process (MRP): the basic tasks on one page.

Section 6 runs the MRP on Austria with this discipline, and Appendix B collects the whole recipe — inputs, equations in order, thresholds, outputs — on two pages.

4.1 Block A: the fundamentals block (annual) — the country's cliff

Basic tasks

Task A1 (growth moments).

Sample mean and standard deviation of log nominal GDP growth in the borrowing currency over the last $T = 25$ years, from the IMF WEO:

$$\hat{\mu} = \frac{1}{T} \sum_{t=1}^T \ln G_t, \quad \hat{\sigma}^2 = \frac{1}{T-1} \sum_{t=1}^T (\ln G_t - \hat{\mu})^2. \quad (11)$$

Nothing else: no filtering, no forecast blending. The COVID and inflation years stay in the sample — they are information about σ , not outliers.

Task A2 (fiscal capacity $\bar{s}$).

The best sustained fiscal performance in the country's own history: the maximum five-year average primary balance (IMF WEO/Fiscal Monitor series pb_j , share of GDP),

$$\bar{s} = \max_t \frac{1}{5} \sum_{j=t-4}^t pb_j. \quad (12)$$

One number, defensible in one sentence: “the country has demonstrated it can sustain this.”

Task A3 (safe rate R^f).

Euro area: the one-year point of the ECB's euro-area AAA yield curve (published daily; taken as a monthly average). Own-currency sovereigns: the one-year point of the own government curve.

Task A4 (haircut h).

Fixed at 0.30, the center of the historical distribution of restructuring haircuts in the Cruces–Trebesch database (public and updated).

Task A5 (capacity objects).

Compute the fundamentals objects of equation (7) at the risk-adjusted drift $\mu - \theta\sigma$, evaluated at $\theta = \hat{\theta}_t$ (from Task B3) with $\rho_\theta = 1$ — risk-off treated as permanent, the conservative teaching default. Under lognormality (1) the closed forms are, with z_M the unique negative solution of $\sigma(1 - \Phi(z_M)) = \phi(z_M)$ (Φ, ϕ : standard normal cdf and density; a one-line numerical solve),

$$\gamma(\theta) = (1 - \Phi(z_M)) e^{\mu - \theta\sigma + \sigma z_M}, \quad b_M(\theta) = \frac{\bar{s} \gamma(\theta)}{R^f - \gamma(\theta)}, \quad d_M(\theta) = (\bar{s} + b_M(\theta)) e^{\mu - \theta\sigma + \sigma z_M}, \quad (13)$$

and the default-probability function used in the funding curve (4) is, from (7) and (1),

$$P_\theta(d) = \Phi\left(\frac{\ln d - \ln(\bar{s} + b_M(\theta)) - (\mu - \theta\sigma)}{\sigma}\right). \quad (14)$$

Further tasks**Task A1⁺ (modifies $\hat{\mu}, \hat{\sigma}$, the output of A1).**

Shrink the historical mean toward the WEO five-year-ahead forecast (equal weights as default; weights published); publish standard errors for $(\hat{\mu}, \hat{\sigma})$, which feed the parameter bootstrap of Task C1⁺.

Task A2⁺ (modifies $\bar{s}$, the output of A2: estimators, then calibration).

Two parts. (i) *Estimators*. Two additional estimators, median of three taken: the fitted value at high debt of a Bohn (1998) fiscal reaction function $s_t = \alpha + \beta d_{t-1} + \text{controls} + \varepsilon_t$ estimated by OLS on the country’s own history, with the Ghosh et al. (2013) “fiscal fatigue” specification (cubic in d) as a robustness check; and the IMF-SRDSF/Commission-DSM realism benchmarks (top-decile sustained consolidations). (ii) *Calibration (discretionary, category 1 of Table 1)*. $\bar{s}$ is a political-economy object — the maximum surplus the *polity* will bear — so soft information determines where within the range spanned by the estimators the published $\bar{s}$ is set, with written justification. Evidence and bright lines: dispersion of domestic bond holdings (Guembel–Sussman logic: widely dispersed domestic holdings make default politically costly, supporting a high credible $\bar{s}$); institutional quality indicators (e.g., the public Worldwide Governance Indicators); instability of the Bohn regression as evidence of surplus fatigue; a restructuring or selective-default history within 30 years sets $\bar{s}$ at the bottom of the estimator range; the North–Weingast configuration (institutionalized commitment that historically raised capacity) supports the top. Every such choice flows through (13): the audit trail shows exactly how the soft information moved the rating.

Task A3⁺ (extends R^f , the output of A3, from a point to a curve).

Publish the whole safe curve $\{R_m^f\}$ by maturity m ; the capacity objects (13) can be re-discounted off the curve without conceptual change.

Task A4⁺ (turns h , the output of A4, into a band and places the baseline within it: calibration).

Report all downstream objects for $h \in \{0.20, 0.50\}$ alongside the baseline 0.30. The placement of the baseline within the band is the second calibration task of Table 1, because who holds the debt shapes who bears the haircut: (i) *holder politics* — widely dispersed domestic holdings support a low h (expropriating one’s own citizens is politically costly), while a history of selective default supports the upper band; (ii) *doom-loop amplification* — a domestic-bank

share of sovereign debt κ above 25% (IMF Arslanalp–Tsuda dataset, quarterly) combined with thin bank capital (IMF Financial Soundness Indicators) steepens the effective P_θ near the cliff, because writing down bonds held by one’s own levered banks writes down one’s own economy; band-edge computations then use the upper part of the h band, widening $[n_L, n_U]$. The OECD Global Debt Report’s holder tables are the cross-check; the placement and its justification are published.

Task A5⁺ (modifies the continuation capacity inside P_θ , the output of A5).

Replace the MRP’s conservative continuation price $\theta^c = \hat{\theta}_t$ by the estimated reversion $\theta^c = \theta_\infty + \hat{\rho}_\theta(\hat{\theta}_t - \theta_\infty)$, with $\hat{\rho}_\theta$ from Task B3⁺; equivalently, compute the schedule $b_M(\theta^c)$ on a grid and evaluate it at the estimated θ^c . This changes P_θ , hence the funding curve, hence the band edges — typically narrowing the fragile region when risk-off episodes are estimated to be transitory.

4.2 Block B: the state-of-affairs (SOA) block (monthly) — the country’s coordinate

Basic tasks

Task B1 (refinancing need n_t).

The gross financing need at annual frequency, assembled from four published numbers: the debt ratio d_{t-1}^{stock} , the short-term (under-one-year) debt stock b_{t-1}^{ST} , the weighted average maturity M_{t-1} of the remaining medium/long-term (MLT) portfolio, and the headline deficit def_t (all shares of GDP):

$$n_t \approx \underbrace{\frac{d_{t-1}^{stock} - b_{t-1}^{ST}}{M_{t-1}}}_{\text{MLT amortizations}} + \underbrace{b_{t-1}^{ST}}_{\text{bills roll within the year}} + \underbrace{def_t}_{\text{new borrowing}}, \quad (15)$$

cross-checked against the GFN reported in the country’s IMF Article IV tables (which since 2022 implement the Sovereign Risk and Debt Sustainability Framework, SRDSF) or the Commission’s Debt Sustainability Monitor. n_t is observable, not estimated; sources for the four components: WEO/Eurostat (debt ratio, deficit), the national debt-management office or the OECD Global Debt Report (short-term stock, WAM).

Task B2 (borrowing costs).

The monthly 10-year benchmark yield (ECB SDW, FRED, national debt-management offices) and its spread ψ_t over the safe curve of Task A3. The model-implied *default component* of the spread is $\psi \approx hp$ by (3) with $p = P_\theta(d_t)$ from (14) evaluated at the issued face value $d_t \approx n_t R^f$; the difference between the observed spread and this component is attributed to liquidity and technical factors, and its size is itself diagnostic (Section 6). For euro-area members the near-zero intra-union spreads of tranquil periods are the good-equilibrium benchmark.

Task B3 (estimating the market price of distress risk $\hat{\theta}_t$).

The object to be estimated is the θ_t of Section 3: the coefficient on standardized log growth in the pricing kernel — the wedge, in units of growth standard deviations, between the physical growth drift μ and the drift $\mu - \theta_t \sigma$ at which the marginal investor prices default risk. It is a *global* object (the same θ_t prices every country in the panel), it is not directly observable, and the MRP estimates it by a transparent two-step plug-in:

1. *Observable input.* Let V_t be the monthly average of the CBOE VIX index — the standard, freely available gauge of global risk appetite; its logarithm is used because the index is

right-skewed. Standardize it against its own history over a fixed window of $W = 240$ months (20 years):

$$z_t = \frac{\ln V_t - m_V}{s_V}, \quad m_V = \frac{1}{W} \sum_{j=1}^W \ln V_{t-j}, \quad s_V^2 = \frac{1}{W-1} \sum_{j=1}^W (\ln V_{t-j} - m_V)^2. \quad (16)$$

z_t measures, in standard deviations of its own history, how elevated global risk aversion currently is; by construction its historical distribution is approximately standard normal.

2. *Affine map into kernel units.* The estimate is

$$\hat{\theta}_t = \max \{0, \theta_\infty + \sigma_\theta z_t\}, \quad (\theta_\infty, \sigma_\theta) = (0.30, 0.25), \quad (17)$$

where the anchor $\theta_\infty = 0.30$ is the Note's conservative long-run level of the distress-risk price — chosen so that the model's average risk premium is consistent with the Longstaff–Pan–Pedersen–Singleton (LPPS) finding that roughly one-third of the *level* of sovereign spreads is a distress-risk premium — and $\sigma_\theta = 0.25$ scales one standard deviation of the proxy into kernel units, consistent with the Note's calibrated illustrations.

The logic of the estimator: the theory says θ_t is high exactly when the global marginal investor's willingness to bear risk is low; the VIX is the most widely used observable proxy for that willingness; (16)–(17) is the simplest monotone map from the proxy into the units the model requires, with both calibration constants documented and contestable. Because z_t is approximately standard normal, (17) also delivers the factor distribution used in (10): $\theta \sim N(\theta_\infty, \sigma_\theta^2)$, so that

$$\text{Exposure}_{it} = 1 - \Phi \left(\frac{\theta_G(n_{it}) - \theta_\infty}{\sigma_\theta} \right) \quad (\Phi : \text{standard normal cdf}).$$

The estimator is re-run every month: each monthly V_t delivers a new $\hat{\theta}_t$ and hence, through Tasks B4–B5 and Block C, a potential rating action. Its acknowledged weaknesses — the VIX is an equity-market object, the map is calibrated rather than estimated, and ρ_θ is imposed at 1 — are exactly what Task B3⁺ repairs.

Task B4 (band edges and regime classification).

Compute $n_L(\hat{\theta}_t)$, $n_U(\hat{\theta}_t)$ by (6) — i.e., solve $\xi'_\theta(d) = 0$ for the two stationary points of the funding curve (4) evaluated with P_θ from (14) — and classify: **safe** ($n_t < n_L$), **fragile** ($n_L \leq n_t \leq n_U$), **distressed** ($n_t > n_U$). Within the fragile region, identify the prevailing equilibrium from the spread level: good-branch pricing (a few bp to ~ 100 bp) versus bad-branch pricing (spreads approaching the recovery ceiling $R^f h / (1 - h)$); the branches are far apart, so the identification is not delicate.

Task B5 (distances and Exposure).

Solve (8) for the critical risk prices $\theta_G(n_t)$, $\theta_B(n_t)$; compute the distances (9) and the exposure probability (10); and compute the *fiscal exit time*

$$X_{it}^{fisc} = \frac{\max\{n_{it} - n_L(\theta_\infty), 0\}}{\bar{s} - s_{it}}, \quad (18)$$

the years of maximum additional fiscal effort needed to leave the band downward — the country's own-control exit in years of effort, complementing the market-controlled exit priced by Δ_G in factor units.

Further tasks

Task B1⁺ (modifies n_t , the output of B1: better source, finer frequency).

Replace the four-number approximation (15) by the published GFN with full maturity machinery: for EU members, from the DWZ replication repository of the Commission DSA; interpolate within the year using the redemption calendar (national treasuries publish auction calendars); cross-check against the OECD Global Debt Report’s issuance and redemption data.

Task B2⁺ (extends ψ_t , the output of B2, across maturities).

Add the 2-year spread (the model’s short-run object) and the term structure of spreads; the slope of the spread curve informs the market’s perceived persistence of stress and is an input to the estimation of ρ_θ in B3⁺.

Task B3⁺ (replaces the estimator (16)–(17), i.e., the outputs $\hat{\theta}_t$, $(\theta_\infty, \sigma_\theta)$, and the imposed $\rho_\theta = 1$).

Two estimators, published side by side. *Reduced form*: the first principal component of standardized monthly spread *changes* across the rated panel, signed so that risk-off is positive, regressed on global proxies (VIX, US equity returns, high-yield spreads) to verify it is the LPPS global factor; the factor’s persistence $\hat{\rho}_\theta$ from an AR(1) fit, and its stationary distribution estimated rather than imposed (Gaussian AR(1) baseline; t -innovations as a fat-tail check). *Structural*: choose $\hat{\theta}_t$ each month to minimize the distance between model-implied spreads — (3) with (14), given each country’s Block-A parameters and n_t , on the good-equilibrium branch where that is the prevailing regime — and observed spreads: a one-dimensional nonlinear least squares. The structural estimate replaces (17) as the rating input; the reduced-form estimate is the audit; discrepancies between the two flag country-specific repricing that should route through Block A (a fundamentals revision), not Block B. The estimated $\hat{\rho}_\theta$ feeds A5⁺, and the estimated factor distribution replaces $N(\theta_\infty, \sigma_\theta^2)$ in (10), with Exposure then computed at the one-year horizon under the AR(1) dynamics rather than the stationary distribution.

Task B4⁺ (modifies the θ entering country i ’s funding curve: the factor loading β_i ; calibration).

The third calibration task of Table 1. Countries differ in how strongly the same global risk-off move hits their funding curve: foreign private investors exit first in risk-off, while a deep domestic nonbank base is stable. Introduce a country loading $\beta_i > 0$ and evaluate P_θ , the band edges (6), and the critical prices (8) at $\theta_{it} = \beta_i \hat{\theta}_t$ (equivalently, the critical prices in global-factor units rescale by $1/\beta_i$, which moves Δ_G and Exposure for country i only). Baseline $\beta_i = 1$; raised (indicatively toward 1.25) for a top-quartile foreign-private share of the investor base, lowered for a predominantly domestic nonbank base — both read from the IMF’s Arslanalp–Tsuda dataset (public, quarterly), with the OECD Global Debt Report’s holder tables as cross-check. The chosen β_i and its justification are published.

Task B5⁺ (attaches uncertainty to the outputs of B4 and B5: regime, θ_G , θ_B , Exposure).

Carry the Block-A standard errors (A1⁺) through the funding-curve solve by parameter bootstrap, yielding a distribution over band edges and hence the robustness tags of Task C1⁺.

4.3 Block C: the scoring and rating block (monthly)

The theory admits three regimes; the rating scale is built on them, refined by distances to the exits. The design principle: *regime membership determines the rating band; exit distances determine the class within the band; hysteresis governs upgrades out of distress; and there are*

no notches — the class is exactly what the thresholds return. Discretion enters this block only through its two DA tasks (Table 1): DA2, which sets the backstop-eligibility state used inside the classification input, and DA1, the discretionary outlook. All thresholds are stated in probability or effort units and published; they are calibration choices, not judgment, and users may recompute the scale under their own thresholds.

Basic tasks

Task C1 (regime classes).

Each country-month is assigned to **S** (safe), **F** (fragile, good-equilibrium pricing), or **D** (distressed: bad-equilibrium pricing, or $n_t > n_U(\hat{\theta}_t)$), carrying over the Task-B4 classification and branch identification.

Task C2 (rating classes within regimes; contains the DA2 eligibility component).

Within **S**, the score is Exposure_{it} from (10):

$$\text{S1: Exposure} \leq 0.01\%; \quad \text{S2: Exposure} \leq 0.1\%; \quad \text{S3: Exposure} \leq 1\%.$$

Within **F**, the primary score is again Exposure_{it} , supplemented by the fiscal exit time X^{fisc} from (18) (can the country leave the band under its own control?):

$$\text{F1: Exposure} \leq 5\% \text{ and } X^{fisc} \leq 2y; \quad \text{F2: Exposure} \leq 15\%; \quad \text{F3: Exposure} \leq 30\%; \quad \text{F4: Exposure} \leq 45\%.$$

Within **D**, the market-controlled exit reverses direction: the score is the probability of the risk-*on* move that destroys the bad equilibrium, $\mathbb{P}[\theta < \theta_B(n_{it})]$ under the Task-B3 factor distribution, supplemented by whether a good equilibrium exists at all at plausible θ :

$$\text{D1: } \mathbb{P}[\theta < \theta_B] \geq 25\%; \quad \text{D2: } \mathbb{P}[\theta < \theta_B] < 25\%; \quad \text{D3: } n_t > n_U(\theta) \text{ for all } \theta \text{ in the factor's 90\% range.}$$

D3 is the state in which no plausible risk-appetite recovery restores a good equilibrium: default is avoidable only by backstop, program, or restructuring. For communication, the ten classes map to familiar letter territory (S1 $\approx$ AAA/AA+, S2 $\approx$ AA/A+, S3 $\approx$ A/BBB+, F1 $\approx$ BBB, F2 $\approx$ BB, F3 $\approx$ B, F4 $\approx$ CCC+, D1 $\approx$ CCC, D2 $\approx$ CC/C, D3 $\approx$ imminent SD), but the native scale is the ten classes with their probability semantics.

The DA2 component of C2 (backstop eligibility; calibration). DA2 produces one object: an **eligibility state** $e_{it} \in \{\text{eligible}, \text{watch}, \text{ineligible}\}$ for the conditional backstop, from observable bright lines in the MRP — euro area: clean compliance with the EU fiscal framework and a low-risk Commission DSA classification $\Rightarrow$ *eligible*; an open excessive-deficit procedure (EDP) with effective action established $\Rightarrow$ *watch*; non-compliance findings $\Rightarrow$ *ineligible*. The state has exactly two consumers, both documented, and touches nothing else in the process. *Consumer one (here, in C2):* if *eligible*, the Exposure input to the threshold tables is computed with the factor distribution truncated at $\theta_G(n_{it})$ — a credible backstop deletes the bad equilibrium for rating purposes (Section 2), so risk-appetite realizations beyond θ_G are excluded; this can move a country from F to S *through the front door* of the measured score. If *watch* or *ineligible*, no truncation is applied. *Consumer two (in C3 below):* the upgrade gate conditions on the state. In addition, the state is information for the outlook (DA1). There is no notch anywhere in this task.

Task C3 (the upgrade gate: hysteresis).

Downgrades into **D** follow the classification instantly (the repricing is discontinuous; pretending otherwise misinforms). Upgrades out of **D** require the theory's strong condition (Section 3): either $\hat{\theta}_t < \theta_B(n_t)$ observed for three consecutive months, or a credible backstop or program event that deletes B — admissible only if the DA2 state is *eligible* (a Draghi-type announcement satisfies this; a capped or conditional-in-doubt backstop does not — it merely relabels θ_B).

Task C4 (the outlook: task DA1; discretionary).

Output: an outlook in {positive, stable, negative}, published with the rating; it never changes the class. Two inputs. (a) *Model-based trajectory (the MRP bright line)*: the sign of the projected twelve-month change in Δ_G (for S/F countries) or in $\mathbb{P}[\theta < \theta_B]$ (for D countries), computed by applying (15) to announced fiscal plans and the redemption calendar at constant θ — the country’s own trajectory toward or away from its exits, never a forecast of the global factor. (b) *Beyond-model information (discretionary in the sense of Principle 4)*: developments the model does not represent but which bear on the trajectory of the coordinate or of the calibrated inputs — backstop politics, banking-sector stress, shifts in the investor base, electoral or institutional events. Input (b) may strengthen or override input (a), with the justification published; it may not touch the class.

Task C5 (publication).

Each rating action publishes: the parameter vector $(\bar{s}, \hat{\mu}, \hat{\sigma}, R^f, h)$; the state $(n_t, \hat{\theta}_t)$; the derived objects $(z_M, \gamma, b_M, d_M, n_L, n_U, \theta_G, \theta_B, \Delta_G, \Delta_B, \text{Exposure}, X^{fisc})$; the regime and class; the DA2 eligibility state; the calibration-layer justifications where the further tasks A2⁺, A4⁺, B4⁺ were applied; the outlook with its rationale; a sensitivity grid over $(\bar{s}, h, \hat{\theta}_t, n_t)$ — the MRP’s honest substitute for formal uncertainty quantification (Section 6 shows it at work); and the code version. Given these, any user reproduces the rating exactly.

Further tasks**Task C1⁺ (attaches uncertainty to the output of C1: the regime assignment).**

Publish the robustness tag from the B5⁺ bootstrap: the fraction of parameter draws that deliver the same regime, with the tag “edge” for a country within one bootstrap standard error of a band edge.

Task DA1⁺ (refines the model-based input of DA1: the projected n -path).

The full trajectory of n_t implied by announced fiscal plans and the redemption calendar, evaluated against the band edges. The relevant question is not the deficit’s sign but its position relative to the band: consolidation that shrinks n_t below $n_L(\hat{\theta}_t)$ *removes the bad equilibrium entirely* — a discrete, nonlinear payoff invisible to smooth cost-benefit calculus — while consolidation deep in the safe region buys little. The refined path analysis sharpens the outlook; it still does not touch the class.

Task DA2⁺ (refines the bright lines behind the DA2 eligibility state).

The full backstop-eligibility criteria (for the ECB’s transmission-protection instrument: compliance with the EU fiscal framework, absence of severe imbalances, debt sustainability as assessed by the Commission/ESM machinery), replacing the MRP’s EDP/DSA-classification proxies as the basis for the state e_{it} . The state’s two consumers (C2 truncation, C3 gate) are unchanged.

Task C5⁺ (extends the output of C5 by governance rules).

Fixed-calendar rating actions for F-class countries and versioned thresholds (Section 7).

A worked illustration

Take the Note’s calibration ($\mu = 3\%$, $\sigma = 3\%$, $\bar{s} = 4\%$, $R^f = 1.03$, $h = 30\%$, plan $s = 2\%$): by (13) at $\theta = 0$, $\gamma = 0.951$, $b_M = 48.5\%$, $d_M = 50.5\%$ of GDP; by (6), $n_L = 39.4\%$, $n_U = 49.2\%$. A country with $n_t = 37\%$ has critical prices $\theta_B = 0.16$, $\theta_G = 0.83$ by (8): it is safe at $\theta_t = 0$ but fragile for $\theta_t \in [0.16, 0.83]$. Suppose Task B3 delivers $\hat{\theta}_t = 0.30$ with factor distribution $N(0.30, 0.25^2)$: the country is fragile on good-equilibrium pricing; $\Delta_G = 0.53$; Exposure = $\mathbb{P}[\theta > 0.83] \approx 1.7\%$ by (10); with $s_t = 2\%$, (18) gives $X^{fisc} \approx 0.5$ years. Class: F1

(fragile, but a modest tail probability and an exit within its own control). If the same country instead carried $n_t = 45\%$: θ_G falls toward 0.3, Exposure rises above 30%, and no feasible surplus path exits the band quickly: class F4 — *with an unchanged debt ratio being consistent with either configuration*. The example is the methodology’s thesis in miniature: the debt level did not determine the rating; the position relative to the fold did.

5 The monetary pillar: euro-area special treatment and own-currency sovereigns

Euro area (full application). Monetary-union membership is the institutional fact that makes the fragile-region apparatus fully applicable: no unilateral monetization, union-wide R^f , backstop institutionally separate and conditional. Three consequences are built into the pipeline. (i) *Common factor, exact comparability*: all members are priced against the same θ_t , so exit distances are directly comparable and the cross-section identifies the factor well (Task B3⁺’s panel is the euro area first). (ii) *Backstop eligibility as a rating input* (Task DA2): the ECB’s transmission-protection instrument and the ESM’s toolkit are conditional backstops; the conditions are observable (fiscal-framework compliance, DSA classification in the Commission machinery replicated by DWZ). An eligible member’s bad equilibrium is deleted for rating purposes; a member drifting out of eligibility loses that protection exactly when it is most likely to need it — the methodology makes this cliff explicit rather than leaving it to narrative. (iii) *Data chassis*: GFNs, maturity structure, and projected n -paths come from the DWZ replication repository (Task B1⁺), giving the euro area the highest-fidelity SOA block in the rated universe.

Own-currency sovereigns (modified application). For the US, UK, Japan and peers, the central bank can always be the residual buyer, so a belief-driven run cannot force *nominal* default: the constitutional backstop deletes B for nominal claims (the De Grauwe contrast: Spain versus the UK, 2011). The pipeline runs unchanged through Blocks A and B, but the outputs are interpreted differently: (i) the headline rating is anchored on the fundamentals block — the distance of inherited debt to d_M and the trajectory of n_t — because outright refusal to roll over is not the operative risk; (ii) fragile-region metrics (Δ_G , Exposure) are published as *shadow* quantities: they price what the sovereign’s position would be without the monetary backstop, and they become operative to the extent the backstop is politically or institutionally constrained (debt-ceiling episodes; central-bank independence under fiscal-dominance pressure — assessed in the Block-A calibration layer, A2⁺); (iii) the residual channel — inflation as the implicit haircut — is reported as a separate *debasement flag* (based on the gap between nominal capacity and real fiscal effort, and on the central bank’s holdings share from Arslanalp–Tsuda), not folded into the default rating. This keeps the rating’s meaning uniform across tiers: probability of contractual non-payment, with monetary erosion disclosed but not conflated.

6 Proof of concept: the minimal process applied to Austria

This section runs the MRP — basic tasks only — on Austria, as of August 2026. Inputs are rounded to teaching precision; each carries its public source. Every computed number below is produced by a numbered equation of the methodology, cited at the point of use; Table 3 collects the full input-to-output chain, and Appendix B states the same chain generically. The point of the exercise is threefold: the entire process is executable from a handful of published numbers; the resulting rating has explicit probability semantics; and the sensitivity grid shows exactly which assumptions the conclusion rests on.

6.1 Block A: fundamentals

A1. Applying (11) to the WEO log nominal GDP growth series 2001–2025: $\hat{\mu} = 3.4\%$, $\hat{\sigma} = 3.0\%$ (recent years: 3.3–3.7% over 2024–26). *A2.* Applying (12) to the Austrian primary-balance series: the best sustained five-year average is roughly +2% of GDP (achieved in the pre-2009 and pre-pandemic expansions), so $\bar{s} = 2.0\%$. The current primary balance is -2.4% (2026 forecast), so the plan is $s_t = -0.024$ — the gap between plan and capacity is itself informative. *A3.* One-year euro-area AAA rate $\approx 2\%$: $R^f = 1.02$. *A4.* $h = 0.30$. *A5.* By (13) at $\hat{\theta}_t = 0.30$ (from B3 below): z_M solves $\sigma(1 - \Phi(z_M)) = \phi(z_M)$ at $\sigma = 0.030$, giving $z_M = -2.27$; then

$$\gamma(0.30) = (1 - \Phi(-2.27))e^{0.034 - 0.009 + 0.030 \times (-2.27)} = 0.947, \quad b_M = \frac{0.020 \times 0.947}{1.02 - 0.947} = 25.8\%,$$

and $d_M = (0.020 + 0.258)e^{0.025 - 0.068} = 26.7\%$ of GDP.

6.2 Block B: state of affairs

B1. The GFN by (15), from four published numbers — debt ratio $d^{stock} \approx 81\%$ of GDP (2025; rising toward 83%), short-term stock $b^{ST} \approx 2\%$, WAM $M = 11.45$ years, deficit $def = 4.3\%$:

$$n_{2026} \approx \frac{81 - 2}{11.45} + 2 + 4.3 = 6.9 + 2 + 4.3 \approx 13.2\% \text{ of GDP,}$$

rounded up conservatively to the baseline $n_t = 14\%$ (the sensitivity grid spans 12–16%). Cross-checks: the federal issuance programme for 2026 is EUR 43–47bn of bonds plus EUR 2–4bn of bills against 2025 GDP of EUR 514bn ($\approx 9\%$ at the federal level, consistent with a general-government figure several points higher). By the identity (2), the face value due is $d_{t-1}/G_t = n_t + s_t \approx 14 - 2.4 \approx 11.6\%$ of GDP — well below $d_M = 26.7\%$. *B2.* The 10-year spread over the German/AAA curve has been of the order of 50bp in mid-2026. The model-implied default component by (3) is $\psi \approx h P_\theta(d_t)$; evaluating (14) at the issued face value $d_t \approx n_t R^f = 14.3\%$ gives

$$P_\theta(d_t) = \Phi\left(\frac{\ln 0.143 - \ln 0.278 - 0.025}{0.030}\right) = \Phi(-23) \approx 0,$$

so $\psi \approx 0$ bp: the observed 50bp is essentially liquidity and technical factors — consistent with, not contrary to, a safe classification. *B3.* By (16), the monthly VIX average in August 2026 stands near its 20-year log mean, so $z_t \approx 0$; by (17), $\hat{\theta}_t \approx \theta_\infty = 0.30$, with factor distribution $N(0.30, 0.25^2)$. *B4.* By (6), solving $\xi'_\theta(d) = 0$ at the Block-A objects:

$$n_L(0.30) = 21.0\%, \quad n_U(0.30) = 26.2\%, \quad n_t = 14\% < n_L \Rightarrow \text{safe},$$

with headroom $n_L - n_t \approx 7.0\%$ of GDP. *B5.* Inverting the edges by (8): $\theta_G(0.14) = 2.33$ and $\theta_B(0.14) = 1.47$. By (9), $\Delta_G = 2.33 - 0.30 = 2.03$ — risk appetite would have to deteriorate by more than eight standard deviations of the factor before Austria’s good equilibrium is threatened — and no bad equilibrium exists at any plausible θ . By (10),

$$\text{Exposure} = 1 - \Phi\left(\frac{2.33 - 0.30}{0.25}\right) = 1 - \Phi(8.1) \sim 10^{-16} < 0.01\%,$$

and by (18), $X^{fisc} = \max\{0.14 - 0.21, 0\}/(0.020 + 0.024) = 0$: Austria is not in the band and needs no effort to stay out of it at current coordinates.

Object	Defined in	Inputs (source)	Austria
$\hat{\mu}, \hat{\sigma}$	eq. (11)	WEO nominal GDP, 2001–2025	3.4%, 3.0%
$\bar{s}$	eq. (12)	WEO/FM primary balances	2.0%
R^f	Task A3	ECB AAA curve, 1y point	1.02
h	Task A4	Cruces–Trebesch center	0.30
$z_t; \hat{\theta}_t$	eqs. (16), (17)	VIX monthly avg. vs. 20y history	≈ 0 ; 0.30
z_M	eq. (13)	$\hat{\sigma}$	−2.27
$\gamma(\hat{\theta}_t)$	eq. (13)	$\hat{\mu}, \hat{\sigma}, \hat{\theta}_t$	0.947
b_M, d_M	eq. (13)	$\bar{s}, \gamma, R^f$	25.8%, 26.7%
n_t	eq. (15)	debt 81%, bills 2%, WAM 11.45y, deficit 4.3% (OeBFA/WEO)	$\approx 14\%$
d_{t-1}/G_t	eq. (2)	$n_t + s_t$	11.6%
$P_\theta(d_t)$; model ψ	eqs. (14), (3)	$d_t \approx n_t R^f$; Block-A objects	≈ 0 ; ≈ 0 bp
n_L, n_U	eq. (6)	funding curve (4) at Block-A objects	21.0%, 26.2%
regime	Tasks B4/C1	n_t vs. $[n_L, n_U]$	safe
θ_G, θ_B	eq. (8)	invert edges at n_t	2.33, 1.47
Δ_G	eq. (9)	$\theta_G - \hat{\theta}_t$	2.03
Exposure	eq. (10)	θ_G ; $N(0.3, 0.25^2)$	$< 0.01\%$
X^{fisc}	eq. (18)	$n_t, n_L(\theta_\infty), \bar{s}, s_t$	0
class	Task C2	Exposure threshold table	S1
eligibility state	DA2 (in C2)	EDP status (Council, July 2025)	watch (no cap)
outlook	Task C4 (DA1)	n -path direction; EDP	negative

Table 3: The Austria case study as an audit trail: object, defining equation, inputs, value.

6.3 Block C: scoring and the rating

C1. Regime **S** (from B4), good-equilibrium pricing. *C2.* Exposure $< 0.01\%$: class **S1** (letter territory AAA/AA+). The DA2 component: the Council opened an excessive deficit procedure for Austria on 8 July 2025, and effective action has subsequently been assessed as taken — eligibility state **watch**, so no truncation of the factor distribution is applied (the cap would be available under clean compliance; at Exposure $< 0.01\%$ it is also not needed). *C3.* The upgrade gate is not engaged: Austria is not in **D**. *C4, the outlook (DA1).* The model-based input is negative: deficits of 4.6–4.3% and a debt ratio rising toward the mid-80s push the projected n -path *up*, shrinking Δ_G , though the path does not enter the band within five years. The beyond-model input reinforces it: the EDP marks the loss of unambiguous backstop eligibility, precisely the state DA2 records as *watch*. Outlook: **negative** — not because of today’s coordinate, which is comfortable, but because both determinants of the coordinate are moving the wrong way; in the methodology’s language, Δ_G is large and shrinking. For completeness: the calibration screens that could move inputs elsewhere in the pipeline do not trigger for Austria — no restructuring history and top-quartile governance ($\bar{s}$ stays at its A2 estimate), domestic-bank share below 25% (h stays at 0.30), and a broad, stable euro-area investor base with the 11.45-year WAM blunting the flight channel (the B4+ loading would stay at $\beta = 1$). *C5.* The publication sheet is Table 3 plus the sensitivity grid below.

6.4 Traceability: from published numbers to the rating

Table 3 restates the case study as an audit trail: every row is one object of the methodology, the equation that defines it, the inputs it consumes, and its Austrian value. Reading the table top to bottom *is* the MRP; Appendix B gives the same sequence without the Austrian numbers.

6.5 The sensitivity grid

The MRP’s substitute for formal uncertainty quantification is a full grid over the four contestable inputs: $\bar{s} \in \{1.5, 2.0, 2.5\}\%$, $h \in \{0.20, 0.30, 0.50\}$, $\hat{\theta}_t \in \{0, 0.3, 0.6\}$, $n_t \in \{12, 14, 16\}\%$ — 81 corners. Austria is **safe in 66 of 81**. Every fragile corner involves $\bar{s} = 1.5\%$ or $h = 0.50$; none occurs at the baseline pair $(\bar{s}, h) = (2.0\%, 0.30)$. The worst examined corner ($\bar{s} = 1.5\%$, $h = 0.50$, $n_t = 16\%$) yields $\theta_G = 0.83$ and, by (10), $\text{Exposure} = 1 - \Phi((0.83 - 0.30)/0.25) \approx 1.8\%$ — i.e., even under jointly pessimistic fiscal capacity and haircut assumptions, with a deteriorated financing need, the tail probability of losing the good equilibrium within the factor’s stationary distribution is about 2% (class F1 territory). The grid thus delivers the teaching conclusion in one table: *Austria’s S1 does not rest on today’s risk appetite ($\hat{\theta}_t$ hardly matters on the grid); it rests on $\bar{s}$ and on keeping n_t low* — which is precisely where Austrian fiscal policy is currently being tested.

6.6 Two remarks the example teaches

The maturity structure is the positioning. Austria’s debt ratio ($\approx 83\%$ of GDP) is three times the model’s maximum face value d_M (26.7%) — and this comparison is a category error that the example is designed to expose. With an 11.45-year WAM, only about a eleventh of the stock rolls in any year: the object the market must refinance, $n_t \approx 14\%$, is what enters the state, and it sits far below the band. The long WAM *is* Austria’s principal risk-off positioning asset: it divides rollover exposure by eleven. A hypothetical shortening of the WAM to 7 years would add roughly 3 points of GDP to n_t at unchanged debt (by (15)) — still safe, but with headroom visibly reduced. Conversely, no plausible improvement in risk appetite would compensate for a WAM that shortened while deficits stayed above 4%. Rating agencies list Austria’s maturity profile as a strength; the MRP prices it.

What the further tasks would change. A2⁺ (Bohn regression on Austrian data) would discipline the contestable $\bar{s} = 2\%$ — the grid shows this is *the* load-bearing assumption; B1⁺ (DWZ repository) would replace the back-of-envelope n_t with the Commission’s maturity machinery; B3⁺ would replace the calibrated map (17) with estimated factor dynamics, turning $\text{Exposure} < 0.01\%$ from a calibration statement into an estimate with a standard error; C1⁺ would convert the 66-of-81 grid count into a formal robustness tag. None of these would change the qualitative conclusion at current coordinates; all of them would sharpen the negative outlook’s quantification.

7 Validation, governance, limitations

Validation. Three tests, run annually and published. (i) *Event coherence*: over 2010–12 and 2020, classified regimes and branch identifications must reproduce the known sequence (near-zero intra-EMU spreads pre-2008 as good-equilibrium selection; Greece/Portugal 2011–12 on the expensive-but-priced bad branch; the 2012 announcement effect as *B*-deletion without purchases). (ii) *Cross-model agreement*: the methodology’s rollover-risk share (the fraction of a country’s spread attributable to θ_t at fixed fundamentals) is correlated with the Bocola–Dovis decomposition where available; persistent disagreement triggers review of Task B3⁺. (iii) *Standards benchmarking*: ratings are compared to SRDSF risk signals, DSM S0 flags, and Manasse–Roubini zones (Appendix A); disagreements are documented with the coordinate that explains them — the methodology should disagree exactly where multiplicity or factor-sensitivity is the story, and nowhere else.

Governance. The parameter vector, code, and thresholds are versioned; threshold changes require a published methodology note with recomputed histories (no silent recalibration). Because

the rating can act as a coordination device inside the fragile band (a downgrade can shift selection toward B), rating actions for F-class countries follow a fixed calendar rather than event-driven timing (Task C5⁺), and the published rationale always distinguishes coordinate changes from repricing — the methodology’s answer to the “ratings as sunspots” concern is procedural discipline plus full transparency, not silence.

Limitations. The model is ability-to-pay with excusable default; strategic default is absorbed into the political-economy calibration of $\bar{s}$ and h (Tasks A2⁺, A4⁺) rather than modeled. One-period debt overstates the speed of crises; the maturity machinery of the DWZ chassis converts the model’s repricing into realistic slow-then-sudden pass-through, but long-maturity sovereigns’ slow-crisis dynamics (Lorenzoni–Werning) deserve a future extension. The factor structure is one-dimensional; regional risk-off (e.g., euro-area-specific stress) is currently absorbed into country residuals and flagged by the B3⁺ audit. Fat tails in θ are handled by the t -innovation check of B3⁺, not by a full extreme-value treatment. None of these limitations is hidden by the scale: every rating carries its sensitivity grid (or, with C1⁺, its robustness tag) and sensitivity bands.

A Why this merits a distinct approach: relation to the standards

We define “distinct” against a benchmark set drawn from international organizations and academic publications (deliberately excluding the methodology documents of the major rating agencies). The set: (i) Cantor–Packer (1996); (ii) IMF Article IV surveillance practice, which since 2022 implements the Sovereign Risk and Debt Sustainability Framework (SRDSF); (iii) the EU’s DSA as described by Darvas–Welslau–Zettelmeyer (2024, “DWZ”); (iv) the European Commission’s Debt Sustainability Monitor (DSM) with its S0/S1/S2 indicators; (v) the OECD Global Debt Report; (vi) IMF debt-at-risk (Fiscal Monitor, October 2024) and its methodological parent, growth-at-risk (Adrian–Boyarchenko–Giannone 2019); (vii) the fiscal-reaction-function tradition: Bohn (1998) and the “fiscal fatigue” debt limits of Ghosh et al. (2013); (viii) Bocola–Dovis (2019); (ix) Manasse–Roubini (2009); (x) the World Bank Fiscal Space Database (Kose et al.). Table 4 states, for each standard, its central risk object, what the risk-off-positioning approach adds, and what we inherit from it.

The table compresses three claims of distinctness that recur across all ten comparisons.

Claim 1: state-dependence and multiplicity. Every standard in the set treats sovereign risk as a (possibly nonlinear) *function* of fundamentals and conditions. The theory says that inside the fragile region it is a *correspondence*: two spreads, one coordinate. This is not a decoration; it changes what a rating is. It implies (a) ratings may legitimately disagree with market prices, in both directions, with a precise statement of when; (b) stress testing must ask not “what if rates rise 1pp” (the DSA’s flat add-on) but “how large a risk-off move destroys the good equilibrium” — a question whose answer, Δ_G , has units and a probability attached; (c) the rating scale needs classes for *regime membership*, not only for risk levels (Task C2).

Claim 2: risk-off positioning as the organizing state variable. The SRDSF conditions on country fundamentals; debt-at-risk conditions on financial conditions; LPPS-style evidence says spreads load on a *global* factor. The methodology promotes that factor to a state variable θ_t with its own dynamics $(\theta_\infty, \rho_\theta)$ and denominates every country’s fragility in its units. This delivers cross-country comparability of a kind no fundamentals checklist can: two countries’ Δ_G are distances along the *same* axis, and Exposure is a genuine probability, not a heuristic index. It also delivers the correct comparative static — a risk-off episode is dangerous in proportion to its perceived permanence — which no standard in the set represents (the DSA’s exogenous-rate path assumes it away; the SRDSF’s stress tests are transitory by construction).

Claim 3: capacity from theory, transparency by construction. Where the standards use estimated thresholds (SRDSF logit cutoffs, S0 signal thresholds, Manasse–Roubini tree splits, Ghosh et al. fatigue limits), the methodology computes its boundaries from an explicit capacity calculation, equation (7), whose inputs are five interpretable parameters. The entire pipeline from data to rating is therefore auditable: change $\bar{s}$ by one point and every downstream object — γ , b_M , d_M , band edges, distances, rating — moves by a computable amount. None of the benchmark standards offers this property; the agencies’ scorecards, which we deliberately exclude, do not either.

We note what we *keep* from the standards with equal emphasis: the GFN concept and its measurement (SRDSF; the 2025 IMF technical note on projecting GFNs), the DWZ replication repository as the euro-area data chassis, the OECD report’s holder tables, Bohn-type regressions for $\bar{s}$, and the DSM’s publication discipline. The methodology is distinct in its organizing principle, not in a rejection of official data practice.

B The minimal rating process at a glance

This appendix summarizes the MRP as a self-contained recipe: the inputs it consumes, the computations in order (each a numbered equation of the main text), the classification rules, and the outputs to publish. It is the generic version of the Austrian audit trail in Table 3; running it for another country means replacing the sources column of that table.

B.1 Inputs: eleven published numbers

Symbol	Object	Source (public, free)
$\{\ln G_t\}_{t=1}^{25}$	log nominal GDP growth series	IMF WEO
$\{pb_t\}$	primary balance series	IMF WEO / Fiscal Monitor
s_t	planned primary balance, current year	IMF WEO forecast / budget
R^f	1-year safe gross rate	ECB AAA curve (euro area); own curve otherwise
h	haircut, fixed at 0.30	Cruces–Trebesch center
d_{t-1}^{stock}	gross debt ratio	WEO / Eurostat
b_{t-1}^{ST}	short-term debt stock	national DMO / OECD GDR
M_{t-1}	weighted average maturity	national DMO / OECD GDR
def_t	headline deficit	WEO forecast
$\{V_{t-j}\}_{j=0}^{240}$	monthly VIX history	CBOE (free)
$(\theta_\infty, \sigma_\theta)$	factor anchor and dispersion, (0.3, 0.25)	calibrated (Note); B3+ estimates them

B.2 Computations, in order

1. **Growth moments:** $\hat{\mu}, \hat{\sigma}$ by (11).
2. **Fiscal capacity:** $\bar{s}$ by (12).
3. **Factor level:** z_t by (16), $\hat{\theta}_t$ by (17); the factor distribution is $N(\theta_\infty, \sigma_\theta^2)$.
4. **Capacity objects:** z_M from $\sigma(1 - \Phi(z_M)) = \phi(z_M)$; then $\gamma(\hat{\theta}_t)$, $b_M(\hat{\theta}_t)$, $d_M(\hat{\theta}_t)$ by (13), and the default-probability function P_θ by (14).
5. **Refinancing need:** n_t by (15); cross-check against the published GFN (SRDSF tables or DSM); face value due $d_{t-1}/G_t = n_t + s_t$ by (2).
6. **Band edges:** $n_L(\hat{\theta}_t)$, $n_U(\hat{\theta}_t)$ by (6), i.e., the proceeds at the two interior stationary points of the funding curve (4) evaluated with P_θ .
7. **Regime:** safe if $n_t < n_L$; fragile if $n_L \leq n_t \leq n_U$ (branch identified from the spread level via (3)); distressed if $n_t > n_U$.

8. **Distances and probabilities:** $\theta_G(n_t), \theta_B(n_t)$ by (8); Δ_G, Δ_B by (9); Exposure by (10); X^{fisc} by (18). For distressed countries, additionally $\mathbb{P}[\theta < \theta_B(n_t)]$, the risk-on exit probability.
9. **Diagnostic spread:** model default component $\psi \approx h P_\theta(d_t)$ at $d_t \approx n_t R^f$, by (3); compare with the observed spread.
10. **Eligibility state (DA2):** {eligible, watch, ineligible} from the bright lines of Task C2; if *eligible*, recompute Exposure with the factor distribution truncated at $\theta_G(n_t)$.
11. **Class:** from the threshold tables of Task C2 (S1–S3 by Exposure; F1–F4 by Exposure and X^{fisc} ; D1–D3 by $\mathbb{P}[\theta < \theta_B]$ and band existence). No notches.
12. **Outlook (DA1):** sign of the projected change in Δ_G (S/F) or in $\mathbb{P}[\theta < \theta_B]$ (D) along the announced n -path at constant θ , adjusted by documented beyond-model information (Task C4).
13. **Sensitivity grid:** recompute steps 4–11 over $\bar{s} \times h \times \hat{\theta}_t \times n_t$ (three values each, 81 corners); report the regime count and the worst-corner Exposure (Task C5).

B.3 Classification thresholds

Class	Regime	Condition
S1 / S2 / S3	safe	Exposure $\leq 0.01\%$ / $\leq 0.1\%$ / $\leq 1\%$
F1	fragile (G)	Exposure $\leq 5\%$ and $X^{fisc} \leq 2y$
F2 / F3 / F4	fragile (G)	Exposure $\leq 15\%$ / $\leq 30\%$ / $> 30\%$
D1 / D2	distressed or (B)	$\mathbb{P}[\theta < \theta_B] \geq 25\%$ / $< 25\%$
D3	distressed	$n_t > n_U(\theta)$ across the factor's 90% range

B.4 Outputs to publish

One page per country: the input vector with sources; the derived objects ($z_M, \gamma, b_M, d_M, n_t, n_L, n_U, \theta_G, \theta_B, \Delta_G, \Delta_B$); regime and class; the DA2 eligibility state; calibration justifications where the further tasks A2⁺/A4⁺/B4⁺ were applied; the outlook with rationale; the 81-corner grid count and worst-corner Exposure; and the code version. Given this sheet, any reader reproduces the rating exactly — which is the property that distinguishes the methodology from every standard in Table 4.

References

- Adrian, T., N. Boyarchenko and D. Giannone (2019), “Vulnerable Growth,” *American Economic Review* 109(4), 1263–1289.
- Arslanalp, S. and T. Tsuda (2014, updated), “Tracking Global Demand for Advanced Economy Sovereign Debt,” *IMF Economic Review* 62, 430–464; dataset updated quarterly.
- Blanchard, O. (2022), *Fiscal Policy under Low Interest Rates*, MIT Press.
- Bocola, L. and A. Dovis (2019), “Self-Fulfilling Debt Crises: A Quantitative Analysis,” *American Economic Review* 109(12), 4343–4377.
- Bohn, H. (1998), “The Behavior of U.S. Public Debt and Deficits,” *Quarterly Journal of Economics* 113(3), 949–963.
- Calvo, G. (1988), “Servicing the Public Debt: The Role of Expectations,” *American Economic Review* 78(4), 647–661.

- Cantor, R. and F. Packer (1996), “Determinants and Impact of Sovereign Credit Ratings,” *FRBNY Economic Policy Review* 2(2), 37–53.
- Cole, H. and T. Kehoe (2000), “Self-Fulfilling Debt Crises,” *Review of Economic Studies* 67(1), 91–116.
- Collard, F., M. Habib and J.-C. Rochet (2015), “Sovereign Debt Sustainability in Advanced Economies,” *Journal of the European Economic Association* 13(3), 381–420.
- Council of the European Union (2025), “Stability and Growth Pact: Council Opens New Excessive Deficit Procedure for Austria,” press release, 8 July 2025.
- Cruces, J. and C. Trebesch (2013), “Sovereign Defaults: The Price of Haircuts,” *American Economic Journal: Macroeconomics* 5(3), 85–117; database updated.
- Darvas, Z., L. Welslau and J. Zettelmeyer (2024), “The Implications of the EU’s New Fiscal Rules,” Bruegel Policy Brief 10/24; replication repository.
- De Grauwe, P. (2012), “The Governance of a Fragile Eurozone,” *Australian Economic Review* 45(3), 255–268.
- European Commission (annual), *Debt Sustainability Monitor*, Institutional Paper series.
- Ghosh, A., J. Kim, E. Mendoza, J. Ostry and M. Qureshi (2013), “Fiscal Fatigue, Fiscal Space and Debt Sustainability in Advanced Economies,” *Economic Journal* 123(566), F4–F30.
- Guembel, A. and O. Sussman (2009), “Sovereign Debt without Default Penalties,” *Review of Economic Studies* 76(4), 1297–1320.
- International Monetary Fund (2022), “Staff Guidance Note on the Sovereign Risk and Debt Sustainability Framework for Market Access Countries,” IMF Policy Paper 2022/039.
- International Monetary Fund (2024), *Fiscal Monitor: Putting a Lid on Public Debt* (debt-at-risk framework), October.
- International Monetary Fund (2025), “A Guide and Tool for Projecting Public Gross Financing Needs,” Technical Notes and Manuals 2025/001.
- Kose, M.A., S. Kurlat, F. Ohnsorge and N. Sugawara (2022), “A Cross-Country Database of Fiscal Space,” *Journal of International Money and Finance* 128.
- Longstaff, F., J. Pan, L. Pedersen and K. Singleton (2011), “How Sovereign Is Sovereign Credit Risk?” *American Economic Journal: Macroeconomics* 3(2), 75–103.
- Lorenzoni, G. and I. Werning (2019), “Slow Moving Debt Crises,” *American Economic Review* 109(9), 3229–3263.
- Manasse, P. and N. Roubini (2009), “‘Rules of Thumb’ for Sovereign Debt Crises,” *Journal of International Economics* 78(2), 192–205.
- OECD (annual), *Global Debt Report*, OECD Publishing.
- OeBFA — Austrian Treasury (2026), *Republic of Austria: Investor Information*, June 2026 (issuance programme, maturity profile, investor base); S&P Global Ratings (2026), Research Update on Austria, 6 February 2026, as republished by OeBFA (used here as a public data compilation for fiscal forecasts, not as a rating standard).
- Stomper, A. (2026), “Sovereign Debt Pricing with Long-Run Capacity and Short-Run Risk Appetite: A Minimal Equilibrium Model,” model note, HU Berlin (“the Note”).

Standard	Its risk object	What positioning adds	What we inherit
Cantor–Packer (1996)	Ratings as regression functions of fundamentals	State-dependence: inside the fragile band the map from fundamentals to risk is a correspondence, not a function; a rating must carry the coordinate (n_t, θ_t) , not only fundamentals	The discipline that ratings be reproducible functions of published observables
IMF Article IV / SRDSF (2022)	Logit crisis probability (near term); fan charts + GFN financeability module (medium term)	Equilibrium multiplicity: GFN <i>thresholds</i> become model-derived band edges $n_L(\theta), n_U(\theta)$ that move with risk appetite; crisis probability becomes a structural tail probability (Exposure)	GFN as the short-run state variable; realism tools for $\bar{s}$; the probabilistic framing
EU DSA (DWZ 2024)	Debt-ratio projections under exogenous market rates	Rates are equilibrium objects; the exogenous-rate convention is an invisible permanent selection of the good equilibrium, so the DSA cannot represent 2010–12 by construction	The data chassis: financing needs, maturity machinery, the replication repository
Commission DSM (S0/S1/S2)	Scored composite early-warning indicator + sustainability gaps	Weights and thresholds derived from theory (distances to band edges), not from signal-extraction on past crises; scores have equilibrium semantics	Publication cadence; the principle of a small transparent indicator suite
OECD Global Debt Report	Stocks, issuance, holder composition, the refinancing wall	Turns descriptive refinancing and holder data into positioning metrics: the wall enters as n_t , holders as amplification parameters (κ , foreign share)	Denominators and holder tables; issuance calendars
IMF debt-at-risk; growth-at-risk (ABG 2019)	Conditional quantiles of debt/growth given financial conditions	A fold instead of smooth quantiles: the conditional distribution is bimodal inside the band; tail risk concentrates at a computable boundary $\theta_G(n)$	Conditioning on a global financial factor; distribution-level reporting
Bohn (1998); Ghosh et al. (2013)	Fiscal reaction functions; fiscal-fatigue debt limits	The limit is forward-looking capacity with priced default (d_M from γ), not estimated past behavior; and there are <i>two</i> relevant boundaries (band edges), not one	The fiscal-reaction regression as an estimator and cross-check for $\bar{s}$
Bocola–Dovis (2019)	Structural decomposition of spreads: fundamental vs. rollover	An implementable geometry (two distances, one tail probability) in place of a full estimated quantitative model; monthly updatable	The validation target: our rollover share should correlate with theirs
Manasse–Roubini (2009)	Crisis “rules of thumb” from classification trees	Risk zones derived from an equilibrium model rather than data mining; zone boundaries move with a state variable	The insight that sovereign risk is zonal, not linear
WB Fiscal Space Database	Cross-country panel of fiscal space indicators	Indicators are <i>combined by the model</i> (into b_M, d_M , distances) rather than juxtaposed	Data breadth and definitional care